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CHENNAI-602105

Engineer to Excel

TECHNICAL QUESTIONS FOR KAAR TECNOLOGIES

Questions for First Round

1. Consider the following C program:

```
#include <stdio.h>
int main()
{
int a[ ] = {2, 4, 6, 8, 10};
int i, sum = 0, *b = a + 4;
for (i = 0; i < 5; i++)
sum = sum + (*b – i) – *(b – i);
printf (“%d\n”, sum);
return 0;
}
```

The output of the above C program is _____

a. 10
b. 12
c. 15
d. 20

Answer (a)

2. Consider the following C program:

```
#include<stdio.h>
 void fun1(char *s1, char *s2){
    char *tmp;
    tmp = s1;
    s1 = s2;
    s2 = tmp;
}
void fun2(char **s1, char **s2){
    char *tmp;
    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}
int main(){
   char *str1 = “Hi”, *str2 = “Bye”;
   fun1(str1, str2); printf(“%s %s “, str1, str2);
   fun2(&str1, &str2); printf(“%s %s”, str1, str2);
   return 0;
}
```

The output of the program above is

a. Hi Bye Bye Hi

b. Hi Bye Hi Bye
c. Bye Hi Hi Bye
d. Bye Hi Bye Hi

Answer (a)

3. Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
   unsigned long int i, j = 0, sum = 0;
   for (i = n; i > 1; i = i/2) j++;
   for ( ; j > 1; j = j/2) sum++;
   return(sum);
}
```

The value returned when we call fun with the input 240 is

a. 4
b. 5
c. 6
d. 40

Answer (a)

4. Consider the following C program

```
#include <stdio.h>
#include <string.h>

void printlength (char *s, char *t) {
  unsigned int c=0;
  int len = ((strlen(s) – strlen(t)) > c) ? strlen(s): strlen(t);
  printf("%d\n",len);
}

void main () {
  char *x = "abc",
  char *y = "defgh";
  printlength (x,y);
}
```

Recall that strlen is defined in string. h as returning a value of type size_t, which is an unsigned int. The output of the program is _____

a. 3
b. 5
c. 7
d. 9

Answer (a)

5. The output of executing the following C program is _____ .

```
#include <stdio.h>
 int total (int v) {
  static int count = 0;
  while(v) {
    count += v&1;
    v >>= 1;
```

```
    }
    return count ;
}

void main () {
    static int x = 0;
    int i =5;
    for (; i>0;i–) {
        x=x+total(i);
    }
    printf(“%d\n”,x);
}
```

a. 23
b. 25
c. 27
d. 29

Answer (a)

6. Consider the following function implemented in C:

```
void printxy(int x, int y) {
    int *ptr;
    x=0;
    ptr=&x;
    y=*ptr;
    *ptr=1;
    printf(“%d, %d”, x, y);
}
```

The output of invoking printxy(1,1) is

a. 0, 0
b. 0, 1
c. 1, 0
d. 1, 1

Answer (c)

7. Consider the C program fragment below which is meant to divide x by y using repeated subtraction. The variables x, y, q and r are all unsigned int.

```
while (r >=y) {
  r = r – y;
  q = q + 1;
}
```

Which of the following condition on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition x == (y*q + r)?

a. (q == r) && (r == 0)
b. (x > 0) && (r == x) && (y > 0)
c. (q == 0) && (r == x) && (y > 0)
d. (q == 0) && (y > 0)

Answer (c)

8. Consider the following snippet of a C program. Assume that swap(&x, &y) exchanges the contents of x and y.

```
int main () {
    int array[] = {3,5,1,4,6,2};
    int done = 0;
    int i;

    while  (done == 0) {
        done = 1;
        for (i = 0, i<=4; i++) {
            if (array[i]< array[i+1]) {
                swap (&array[i], &array[i+1]);
                done = 0;
            }
        }
        for (i=5; i>=1; i–) {
            if (array[i] > array [i-1]) {
                swap(&array[i], &array[i-1]);
                done = 0;
            }
        }
        printf(“%d”, array[3]);
}
```

The output of the program is _____

a. 3
b. 4
c. 5
d. 6

Answer (a)

9. Consider the following C program.

```
#include<stdio.h>
int main () {
        int m = 10;
        int n, nl;
        n = ++m;
        n1 = m++;
        n–;
        –n1;
        n-= n1;
        printf (“%d”,n),
        return 0;
}
```

The output of the program is _____

a. 0
b. 1
c. 5
d. None of the above

Answer (a)

10. What will be the output of the following C program?

```
void count(int n){
  static int d=1;
     printf("%d ", n);
     printf("%d ", d);
     d++;
     if(n>1) count(n-1);
     printf("%d ", d);
  }

  void main(){
     count(3);
  }
```

a. 3 1 2 2 1 3 4 4 4
b. 3 1 2 1 1 1 2 2 2
c. 3 1 2 2 1 3 4
d. 3 1 2 1 1 1 2

Answer (a)

11. The number of tokens in the following C statement is?

```
printf("i = %d, &i = %x", i, &i);
```

(a) 3 **(b)** 26 (c) 10 **(d)** 21

Answer: (c)

12. What will be the output of the following C program segment?

```
char inchar = 'A';
switch (inchar)
{
case 'A' :
printf ("choice A \n") ;
case 'B' :
printf ("choice B ") ;
case 'C' :
case 'D' :
case 'E' :
default:
printf ("No Choice") ;
}
```

(a) No choice
(b) Choice A
(c) Choice A Choice B No choice
(d) Program gives no output as it is erroneous

Answer: (c)

13. Consider the following C function definition:

```
int Trial (int a, int b, int c)
{
if ((a > = b) && (c < b)) return b;
else if (a > = b) return Trial (a,c,b);
else return Trial (b,a,c);
}
```

The function Trial:

(a) Finds the maximum of a, b, and c
(b) Finds the minimum of a, b and c
(c) Finds the middle number of a, b, c
(d) None of the above

Answer: (c)

14. The value of j at the end of the execution of the following C program.

```
int incr (int i)
{
static int count = 0;
count = count + i;
return (count);
}
main ()
{
int i,j;
for (i = 0; i <=4; i++)
j = incr(i);
}
```

(a) 10 (b) 4 (c)6 (d) 7

Answer: (a)

15. Consider the following declaration of a ‘two-dimensional array in C:

```
char a[100][100];
```

Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0, the address of a[40][50] is
(a) 4040 (b) 4050 (c) 5040 (d) 5050

Answer: (b)

16. Consider the following C-program:

```
void foo(int n, int sum)
{
int k = 0, j = 0;
if (n == 0) return;
k = n % 10; j = n / 10;
sum = sum + k;
foo (j, sum);
printf ("%d,", k);
}
int main ()
{
int a = 2048, sum = 0;
foo (a, sum);
printf ("%d\n", sum);
getchar();
}
```

What does the above program print?
(a) 8, 4, 0, 2, 14
(b) 8, 4, 0, 2, 0
(c) 2, 0, 4, 8, 14
(d) 2, 0, 4, 8, 0

Answer: (d)

17. Consider the following C function:

```
int f(int n)
{
```

```
static int i = 1;
if (n >= 5)
return n;
n = n+i;
i++;
return f(n);
}
```

The value returned by f(1) is

(a) 5 (b) 6 (c) 7 (d) 8

Answer: (c)

18. Consider the following C program

```
void main()
{
int x, y, m, n;
scanf ("%d %d", &x, &y);
/* x > 0 and y > 0 */
m = x; n = y;
while (m != n)
{
if(m>n)
m = m - n;
else
n = n - m;
}
printf("%d", n);
}
```

The program computes

(a) x + y using repeated subtraction

(b) x mod y using repeated subtraction

(c) the greatest common divisor of x & y

(d) the least common multiple of x & y

Answer: (c)

19. Consider the following C-program:

```
double foo (double); /* Line 1 */
int main () {
double da, db;
// input da
db = foo (da);
}
double foo (double a) {
return a;
}
```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

(a) no compile warning or error

(b) some compiler-warnings not leading to unintended results

(c) some compiler-warnings due to type-mismatch eventually leading to unintended results

(d) compiler errors

Answer: (d)

20. Consider line number 3 of the following C-program.

```
int main ( ) { /* Line 1 */
int i, n; /* Line 2 */
fro (i =0, i<n, i++); /* Line 3 */
}
```

Identify the compiler's response about this line while creating the object-module:

(a) No compilation error
(b) Only a lexical error
(c) Only syntactic errors
(d) Both lexical and syntactic errors

Answer: (c)

ADVANCED CODING QUESTIONS ROUND 2

1. Given an array arr[] of N elements, sort the array according to the following relations:
   arr[i] >= arr[i – 1], if i is even, ∀ 1 <= i < N
   arr[i] <= arr[i – 1], if i is odd, ∀ 1 <= i < N
   Print the resultant array.
   Examples:
   Input: N = 4, arr[] = {1, 2, 2, 1}
   Output: 2 1 2 1
2. Given an array arr[] of integers, segregate even and odd numbers in the array such that all the even numbers should be present first, and then the odd numbers.
   Examples:
   Input: arr[] = {7, 2, 9, 4, 6, 1, 3, 8, 5}
   Output: 2 4 6 8 7 9 1 3 5

   Input: arr[] = {1, 3, 2, 4, 7, 6, 9, 10}
   Output: 2 4 6 10 7 1 9 3

3. The Next greater Element for an element x is the first greater element on the right side of x in the array. For elements for which no greater element exist, consider the next greater element as -1.
   Example:
   Input: arr[] = [4 , 5 , 2 , 25]
   Output: 4 –> 5
   5 –> 25
   2 –> 25
   25 –> -1
   Explanation: except 25 every element has an element greater than them present on the right side
   Input: arr[] = [13 , 7, 6 , 12]
   Output: 13 –> -1
   7 –> 12
   6 –> 12
   12 –> -1
   Explanation: 13 and 12 don't have any element greater than them present on the right side

4. Given an array, for each element find the value of the nearest element to the right which is having a frequency greater than that of the current element. If there does not exist an answer for a position, then make the value ‘-1’.
   Examples:

   Input: a[] = [1, 1, 2, 3, 4, 2, 1]
   Output : [-1, -1, 1, 2, 2, 1, -1]
   Explanation:
   Given array a[] = [1, 1, 2, 3, 4, 2, 1]
   The frequency of each element is: 3, 3, 2, 1, 1, 2, 3.